

Gradient Adjusted Pace (GAP)

Step 1. Get the slope (gradient)

Formula:

$$\text{slope} = \Delta h / d$$

where:

- Δh = elevation gain/loss (meters)
- d = horizontal distance covered (meters)

Slope percentage:

$$\text{slope \%} = (\Delta h / d) \times 100$$

Step 2. Estimate energy cost of running on that slope

Research (Minetti et al., 2002) shows the oxygen cost of running depends on slope.

Key points:

- Uphill → adds effort, equivalent to running faster on flat
- Downhill → reduces effort (up to a point, ~ -5% grade is optimal), but steep downhills increase braking cost

Simple model:

$$C(\text{slope}) = C_0 \times (1 + k \times \text{slope})$$

where:

- C_0 = cost on flat ground
- $k \approx 30\text{--}50$ depending on source

Step 3. Adjust your pace

Formula:

$$\text{GAP} = \text{actual pace} \times (C(\text{slope}) / C(0))$$

Example:

- Running 5:00 min/km up a 5% hill (slope = 0.05)
- Assume $k = 30 \rightarrow$ energy cost = $1 + 30 \times 0.05 = 2.5$ times higher
- $\text{GAP} \approx 5:00 \div 2.5 \approx 2:00$ min/km (flat equivalent)

Note: Real models are nonlinear and moderated for realism.

Step 4. Apply across an entire run

Process:

1. Break the run into small segments (e.g., 10–100 m).
2. Calculate slope and GAP for each segment.
3. Average the GAP over the whole distance/time.

In practice

- Strava, TrainingPeaks, Stryd, and Garmin all implement proprietary GAP/Effort Pace models.
- They are based on the Minetti curve for running energetics, which looks like a U-shape: effort increases both uphill and downhill.